

BOUREMA DIARRA

DATA ANALYST | PYTHON | SQL | POWER BI

PROFESSIONAL SUMMARY

Data Analyst and environmental researcher with practical experience working with real-world data, from cleaning and organising datasets to analysing patterns and communicating results. I work with Python (Pandas, NumPy, Jupyter), Excel, statistical analysis and data visualisation, while continuing to strengthen my skills in SQL and Microsoft Power BI. My research experience has taught me to work carefully with complex quantitative and qualitative data, question results, identify data-quality issues, and turn findings into clear reports and visual outputs. I bring curiosity, attention to detail and an independent approach to solving data problems, together with experience working across international and multicultural environments.

TECHNICAL SKILLS

Data Analysis & Programming: Python (Pandas, NumPy, Jupyter), Excel, R, SPSS, exploratory data analysis, data cleaning, data processing, data validation, statistical analysis

Databases & Business Intelligence: SQL (developing), Microsoft Power BI (developing), Power Query, data modelling, DAX fundamentals

Additional Analytics: Data visualisation, web scraping, text mining, machine-learning model development and evaluation

Spatial Analysis: ArcGIS

SELECTED DATA ANALYTICS PROJECTS

Python Data Wrangling & Preprocessing

GitHub Portfolio

- Turned a messy, real-world laptop dataset into clean, analysis-ready data using Pandas, resolving missing values, inconsistent data types and quality issues while applying normalisation, feature engineering and categorical encoding.
- Built a clear, reproducible preprocessing workflow that transformed raw data into a reliable foundation for further analysis and modelling.

Financial Data Collection & Web Scraping Pipeline

GitHub Portfolio

- Built a Python web-scraping workflow that turned raw financial information from web sources into clean, structured and analysis-ready data through validation, transformation and quality checks.
- Explored and visualised the prepared data to uncover meaningful patterns and communicate the findings clearly, documenting the full process in a reproducible Jupyter Notebook.

Automobile Price Analysis & Predictive Modelling

GitHub Portfolio

- Explored automobile pricing data with Python, Pandas and NumPy to uncover the factors associated with price differences, using visualisation, correlation analysis and statistical testing to investigate patterns in the data.
- Built and evaluated regression models to predict automobile prices, refining features and comparing model performance to improve the reliability of the predictions.

PROFESSIONAL EXPERIENCE

PhD Researcher

Eötvös Loránd University | Budapest, Hungary | 09/2023 - Present

- Turn complex qualitative and quantitative research data into clean, accurate and well-structured datasets, resolving quality issues and maintaining clear documentation throughout the research process.
- Explore environmental and socioeconomic data to uncover meaningful patterns, understand what is driving the results, and turn the findings into clear, evidence-based insights.
- Lead an independent research project on fuelwood use and land degradation in Mali, taking it from study design and field data collection through analysis, reporting and communication of findings.
- Conducted 85+ semi-structured interviews with households, community representatives, local leaders and forestry officials, building first-hand evidence while coordinating fieldwork across multiple municipalities.
- Turn complex analytical findings into clear, accessible reports and presentations, helping different audiences understand the key results and their implications.

Research Intern

United Nations Framework Convention on Climate Change (UNFCCC) | Remote | 01/2024 - 07/2024

- Reviewed, organised, and quality-checked information and documentation for international teams, improving consistency, clarity, and accuracy before delivery.
- Prepared reports, background materials, and written communications for international audiences while managing multiple priorities in a remote environment.

Research Intern

United Nations Convention to Combat Desertification (UNCCD) | Remote | 01/2022 - 07/2022

- Collected, reviewed, validated, and organised information for publication, maintaining high standards of data and information quality.
- Independently led the “Children vs UNCCD” initiative from idea to implementation, working with local communities to bring the project to life and plant 50 fruit trees across two villages in Mali.

Teacher of Geography and History

High School Ramata Touré | Bamako, Mali | 11/2021 - 08/2023

- Planned and delivered geography and history instruction across three grade levels, translating complex topics into accessible learning materials and assessing student progress throughout the learning process.

EDUCATION

PhD Candidate in Environmental Science - Eötvös Loránd University, Budapest, Hungary | Expected August 2027

MSc in Ecology and Nature Conservation - Perm State National Research University, Perm, Russia | 2018 - 2021 | GPA: 4.61/5

Bachelor of Teaching in History and Geography - Normal Higher School of Bamako, Mali | 2013 - 2015

Bachelor's Degree in Geography and Planning - University of Social Sciences and Management of Bamako, Mali | 2008 - 2013

LANGUAGES AND ADDITIONAL INFORMATION

Languages: French (Native) | English (Fluent) | Bambara (Native) | Russian (Intermediate)

Availability: Eligible to work in Hungary with a valid residence permit | Open to remote, hybrid, and onsite opportunities